

GRAPHIC ERA (DEEMED TO BE UNIVERSITY), DEHRADUN

SEMESTER I

Name of Department: - Department of Computer Applications

Course Code:		TMC101		Course Title:		Full Stack Development		
Credits:	3	Semester:		I	Category of Course:		DSC	
TEACHING PERIODS (In hours per week)			TW+SL (in hours per semester)			Relative Weight		
L	T	P	45			CIE	MSE	ESE
3	0	0				25	25	50
Examination Duration (Hrs):			Theory				Practical	
			3				0	

Pre-requisite: Fundamental knowledge of network, Internet and WWW.

Course Outcome:	After completion of the course the students will be able to: CO1: Describe and demonstrate the usage of XHTML tags and CSS for creating presentable web pages. CO2: Describe the use of JavaScript for creating interactive web pages. CO3: Develop dynamic web pages by applying event-handling mechanisms using JavaScript. CO4: Develop web pages using cookies, sessions, and databases in PHP. CO5: Describe the concepts of the MERN stack used in Full Stack Development and the features and components of ReactJS.
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Details of the Course:

Sl. No.	Contents	Contact Hours
1	Unit 1: Basics of XHTML Standard Structure of XHTML document, Difference between HTML and XHTML, Basic Text Markup Elements – paragraph <p> , heading tags H1..H6, break, italic, bold, superscript, subscript, emphasis, <hr>, Character entities, Hyperlinks , Images Lists , , <dl>, tables <table>, division <div> and span , Form Elements, Label, Text boxes - password, button, submit, reset, Placeholder, Textarea, Radio button, Checkboxes, Button, Dropdown <select>	8

2	Unit 2: Introduction to CSS,CSS Properties – text, font, list, color, Selectors, Simple/Element Selector, Id Selector, Class Selector, Generic Selector, Universal Selector, Pseudo-class Selector, Types and Levels of Style: Inline, Internal, External, CSS BoxModel, Background images Examples using HTML and CSS	8
3	Unit 3: Introduction to JavaScript, Event Handling, DOM, and Dynamic Documents with JavaScript Introduction to JavaScript - Overview of JavaScript, JavaScript Syntax, Variables and Data Types Screen Output and Keyboard Input,alert(),confirm(),prompt(), document.write(),Controls and Loops, String Methods, Arrays, Functions, Pattern Matching. Event Handling, DOM, and Dynamic Documents with JavaScript - Document Object Model (DOM),DOM tree structure of HTML, Accessing elements in JavaScript, Event and Event Handling, Handling events from body elements,Handling events from button elements,Handling events from text boxes,Validations on Forms simple	10
4	Unit 4: PHP Basics and Advance Features PHP Basics – Introduction and basic syntax of PHP, Control Statements with examples, Output Statements,String Functions, Arrays, Functions PHP Advance Features – Form Handling, Cookies and Sessions Management Creating cookies and session variables in PHP programs, Creating a simple database and database operations	9
5	Unit 5: Introduction to MERN – Overview of Full Stack Web Development, Overview of MERN, Overview of MERN Components,ReactJS,Node.js,Express,MongoDB Tools and Libraries, React library Introduction to ReactJS - React features, benefits, and applications, Advantages and disadvantages of ReactJS,ReactJS Vs other Front-End Technologies, ReactJS development environment setup,Creating and executing a new ReactJS project, Folder Structure of ReactJS applications	10
	Total	45

Text Books:

Authors Name	Title	Edition	Publisher, Country	Year
Robert W. Sebesta	Programming the world wide web	6 th Edition	Pearson	2018
Vasan Subramanian	Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node	2 nd Edition	Apress	2020

Reference Books:

Authors Name	Title	Edition	Publisher, Country	Year
Kogent Learning Solutions Inc	HTML 5: Covers CSS3, JavaScript, XML, XHTML, AJAX, PHP & jQuery: Black Book	2 nd Edition	Dreamtech Press.	2016
Eddy Wilson	MERN Quick start guide: Build Web applications with MongoDB, Express.js, React and Node	1 st Edition	Packt publishing.	2018